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Notched Tariffs and Rooftop Solar: How Morocco's Selective Electricity Pricing Shapes the Household Value of Photovoltaic Self-Consumption under Law 82-21

Research Problem

Morocco imports about 90% of the energy it consumes. It also aims for renewables to make up 52% of installed power capacity by 2030 [1]. Buildings, residential and tertiary together, account for about 32% of final energy consumption [2]. A photovoltaic (PV) system in Morocco can produce 1,600–1,900 kWh per installed kWp each year [1], so household rooftop solar is a natural way to serve both goals.

Law No. 82-21 on electricity self-production, promulgated in 2023, waited about three years for its implementing texts. Decree No. 2-25-100 of 5 March 2026, published in Bulletin officiel no. 7489, and Decision No. 04/26 of the National Electricity Regulatory Authority (ANRE) finally completed the framework, and the decree took effect on 9 June 2026 [3], [4], [5]. Installations below 11 kW, the size of a household system, follow a simplified procedure [4], [6], [7].

The new framework, however, values surplus electricity very little. A self-producer may sell at most 20% of annual output to the grid. For the regulation period from 1 March 2026 to 28 February 2027, ANRE sets the price of that surplus at 0.21 Moroccan dirhams (MAD) per kWh in peak hours and 0.18 MAD/kWh off-peak, but this tariff covers only the high-, extra-high- and medium-voltage networks [5], [8]. No standard remuneration has yet been fixed for low-voltage residential exports, and several implementing texts are still pending [2], [9], [7]. Retail electricity for households costs between 0.90 and 1.60 MAD/kWh [10]. A household PV system therefore earns almost all of its value from the grid electricity that the household no longer buys, and that value depends on the design of the retail tariff.

That design is unusual. Consumption up to 150 kWh per month is billed in progressive blocks. Above 150 kWh, the national utility (ONEE) applies “selective” pricing: the whole month's consumption is billed at the price of the block that the household reaches, with a tolerance of 10 kWh per block [11], [10]. Crossing a threshold therefore raises the price of every kilowatt-hour consumed that month, and not just of the last ones. Public economists call this a notch, as opposed to a kink, where only the next kilowatt-hour costs more [12]. Rooftop PV interacts with notches directly, because a panel that lowers monthly grid consumption can push a household below a threshold and reprice its entire bill.

A simple calculation with the official block prices shows how large the effect can be (Table 1 and Fig. 1). Take a household that draws 340 kWh from the grid in a summer month, and pays about 470 MAD for energy. If self-consumed solar electricity removes 40 kWh from that draw, the bill falls to about 350 MAD, which is a saving of 2.99 MAD for each avoided kilowatt-hour. A conventional progressive tariff with identical prices would save only 1.38 MAD/kWh on the same 40 kWh, so the notch multiplies the saving by 2.2. This calculation is only an illustration; the study itself measures the effect properly.

The value of solar self-consumption in Morocco is therefore strongly non-monotonic in household consumption, with sharp peaks just above each threshold. The Moroccan assessments reviewed below model either a single dwelling under the optional time-of-use tariff [2] or PV against unsubsidized prices [13]. If notches matter as much as this calculation suggests, neither the profitability of rooftop solar under the new law nor the question of which households will adopt it can be predicted from average tariffs. That is the problem this research addresses.

Table 1. Illustrative bill impact of 40 kWh of monthly PV self-consumption (author's calculation from the official block prices [10], [11]; fixed charges and taxes excluded).

Grid draw before → after PV (kWh per month)	Selective bill before → after (MAD)	Selective saving (MAD per kWh)	Progressive bill before → after (MAD)	Progressive saving (MAD per kWh)	Ratio
90 → 50	81.09 → 45.05	0.90	81.09 → 45.05	0.90	1.00
175 → 135	187.81 → 127.66	1.50	170.59 → 127.66	1.07	1.40
340 → 300	469.78 → 350.28	2.99	369.45 → 314.18	1.38	2.16

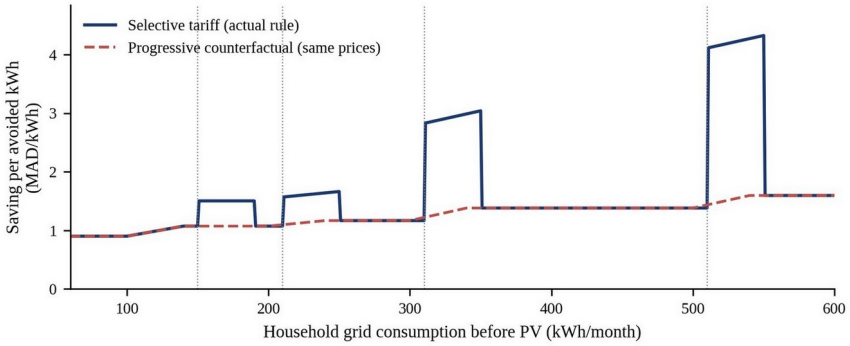


Fig. 1. Illustrative saving per avoided kWh when self-consumed PV lowers monthly grid consumption by 40 kWh, under the actual selective rule and a progressive counterfactual with identical block prices (author's calculation from [10], [11]). Dotted lines mark the 150 kWh threshold and the selective block limits including the 10 kWh tolerance; the progressive counterfactual steps at 100, 200, 300 and 500 kWh.

Existing Literature

Tariff design and the private value of PV. International studies agree that what a household earns from rooftop solar depends on the tariff as much as on the sun. Darghouth et al. [14] found that the bill savings of Californian PV owners vary widely with rate design and with how much electricity the customer uses when prices are tiered. Borenstein [15] studied 2007–2014 data from the US utility with the most residential solar customers. He found that tariff design and net metering shape private incentives, and that adoption leaned toward wealthy households. In a simulation of 448 households, Lorenz et al. [16] found that the gains from PV–battery systems differ between fixed and dynamic tariffs. Sepúlveda-Vélez et al. [17] ran an hourly net present value (NPV) model on 30 Colombian scenarios covering six socioeconomic strata. Their maximized NPV ranged from €2.8 to €2,816, so profitability was very uneven, and they concluded that tariff subsidies can make PV less attractive to low-income users.

Nonlinear prices and notches. Ito [18] showed that households facing nonlinear electricity prices react to the average price rather than the marginal one. Under selective pricing these two prices differ greatly, so the gap between them shapes behaviour. Kleven and Waseem [12] defined notches as discrete jumps in total liability at a threshold. They showed that notches create strong local incentives that differ in kind from those of kinked schedules.

Utility and distributional effects. When prosumers buy fewer kilowatt-hours under volumetric tariffs, fixed network costs fall on customers who do not install PV. Eid et al. [19] and Picciariello et al. [20] measured this transfer from ordinary consumers to prosumers, which is why the question of who benefits from PV also matters for tariff policy.

Morocco. Allam et al. [2] sized PV and batteries together for a single dwelling in Oujda with 14.45 MWh of annual demand, under ONEE's optional time-of-use tariff and the 20% export quota. They note that such households in practice run on pure self-consumption. Kettani and Sanin [13] estimated that 14% of Moroccan households are energy poor. They also found that removing subsidies for households where PV already competes at the unsubsidized price would

free 25% of the budget needed to close the electricity poverty gap. Benbba et al. [1] review the national solar resource and strategy.

Gap. Three gaps follow from this review. First, the Moroccan studies either examine one high-consumption dwelling on an optional tariff [2] or were completed before the 2026 implementing decree [13]. Second, to the author's knowledge, no study has isolated how selective repricing changes the value of PV. Third, the international evidence deals with kinked block tariffs [14], [15] and not with whole-bill notches, so it cannot simply be carried over to Morocco.

Research Question

How much does Morocco's selective residential tariff, which reprices the entire monthly bill once a block threshold is crossed, change the net present value of rooftop solar for households in Casablanca under Law 82-21, compared with a progressive tariff with identical block prices?

Scope and variables. The comparison covers 1–3 kWp systems and household consumption between 50 and 600 kWh/month. The independent variables are tariff design (the actual selective rule versus the progressive counterfactual) and baseline household consumption. The main dependent variable is the 25-year NPV in MAD, complemented by the discounted payback period (years) and the effective value per self-consumed kilowatt-hour (MAD/kWh).

H1 (mechanism): Under selective pricing, the value of a self-consumed kilowatt-hour is non-monotonic in consumption, and even after averaging over the year and over realistic seasonal demand it exceeds the marginal block price by at least 25% for the archetypes closest to a threshold.

H2 (magnitude): Averaged over the archetypes above 150 kWh/month, selective pricing raises the annual bill saving of a 2 kWp system by at least 15% relative to the progressive counterfactual. The largest gains go to households whose seasonal consumption peaks just above a block threshold.

H3 (equity): Households consuming under 100 kWh/month, which pay only the lowest block price, obtain the lowest NPV per installed kWp of all consumption levels. The share of Monte Carlo draws with a positive NPV is lower for them than for any band above 150 kWh/month.

Methodology

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Design. The study is a quantitative, simulation-based techno-economic analysis, and everything rests on one counterfactual comparison. Each household archetype is evaluated twice: once under the actual selective rule, and once under a progressive tariff that has the same block prices but charges each block only at the margin. Since only the billing rule differs, the gap between the two results isolates the effect of selective pricing. That effect works through self-consumed electricity, so it does not depend on the unsettled export rules, which only shift the level of NPV and are bracketed in Step 4. No surveys or interviews are used.

Step 1 – Regulation and tariffs. The study collects Law No. 82-21, Decree No. 2-25-100, ANRE Decision No. 04/26 and the current low-voltage residential schedules of ONEE and of the Casablanca distributor. For monthly consumption C up to 150 kWh, the energy bill is $0.9010 \cdot \min(C, 100) + 1.0732 \cdot \max(0, C - 100)$. For C above 150 kWh, ONEE applies the price of the block reached to “la totalité de la consommation mensuelle” (the whole of the monthly consumption) [11], so the bill is $C \cdot pk$, where pk is that price (blocks of 151–200, 201–300, 301–500 or above 500 kWh, each with the 10 kWh tolerance) [10]. The 150 kWh threshold itself is treated as exact, and the progressive counterfactual uses the same prices without any tolerance. The billing engine also includes the fixed charges and taxes printed on real bills, and it is validated by reproducing at least three real household bills to within 1 MAD. Some of these bills, collected under a confidentiality agreement with the participating households, are already

in hand and confirm that ONEE bills this tariff on a strict monthly cycle, so the thresholds used throughout this study are monthly ones.

Step 2 – PV generation. Hourly output for a typical meteorological year in Casablanca (33.57° N, 7.59° W) comes from the European Commission's PVGIS tool [21]. Casablanca is the reference site because it is the largest city in Morocco and because bills and installer quotes can be collected there in person. The model covers 1, 2 and 3 kWp crystalline-silicon systems facing south at a 30° tilt, with 14% system losses (the assumptions used in [2], [17]) and a degradation of 0.5% per year, the median rate across nearly 2,000 field measurements [22], over a 25-year life (the lifespan used in [2]). Archetypes below 150 kWh/month are evaluated at 1 kWp only, since a larger array there would generate far more than the household can absorb once the 20% export quota is reached. The simulated yield is checked against the 1,600–1,900 kWh/kWp range reported for Morocco [1].

Step 3 – Household demand. Twenty-three archetypes are defined, with mean monthly consumption from 50 to 600 kWh in steps of 25 kWh. Their seasonal pattern is calibrated on twelve-month billing histories from at least 20 Casablanca households across consumption levels, in the stratified spirit of Sepúlveda-Vélez et al. [17], who assessed six socioeconomic strata. Hourly load shapes are needed only to split PV output between self-consumption and export, and a normalized residential profile adapted from [2] is scaled to each archetype. Because the selective mechanism acts on monthly totals, the hourly shape only affects the self-consumption ratio r, the share of monthly PV output used on site, which is varied between 30% and 70% so that the conclusions do not depend on one assumed profile. Self-consumption can never exceed the household's own demand: in month m, SCm = min(r·Gm, Cm), where Gm is the PV output of that month and Cm the household's consumption. Any surplus is exported and is paid only within the 20% annual quota (Step 4).

Step 4 – Bill savings. For each archetype, month and tariff, the saving is B(Cm) − B(Cm − SCm), where B is the billing function and SCm the PV electricity self-consumed that month. Savings are computed on the total bill, whereas Table 1 shows the energy component only. Two export regimes are modelled: in the first, exports earn nothing, and in the second they are paid 0.18–0.21 MAD/kWh up to 20% of annual output [5]. The second regime applies the ANRE medium-voltage tariff by analogy, since no low-voltage tariff exists yet, so it is a proxy and not a legal rule.

Step 5 – Financial model. NPV = −I + Σ (St − OMt) / (1 + d)t, summed over t = 1 to 25. Here I is the upfront cost, St the annual saving (with degradation and a tariff escalation rate g), OMt the operation and maintenance cost (1% of I per year plus an inverter replacement in year 12) and d the discount rate, set at 5%, 7% and 9%. The cost per kWp comes from written quotes from at least five Casablanca installers, cross-checked against published Moroccan cost assumptions [2]. The discounted payback period, the internal rate of return and the effective value per self-consumed kWh (annual saving divided by annual self-consumed kWh) complete the indicators.

Step 6 – Uncertainty and hypothesis testing. A Monte Carlo simulation with 10,000 draws varies capital cost (±20%), discount rate, tariff escalation (0–3% per year), self-consumption ratio and PV yield (±5%), and a bootstrap of the household calibration sample (1,000 resamples) carries its uncertainty into the results. For each archetype it reports the median and the 90% interval of ΔNPV = NPV(selective) − NPV(progressive), together with the probability that ΔNPV > 0. Each hypothesis is judged against its stated threshold. H1 is supported if, in the annual results, the effective value per kWh exceeds the marginal block price by at least 25% for the archetypes closest to a threshold. H2 is supported if the lower bound of the 90% interval of the mean gap in annual saving stays above 15% for the archetypes above 150 kWh/month. H3 is supported if the archetypes under 100 kWh/month have both the lowest median NPV per kWp and the lowest share of positive-NPV draws. The ranking of gains across archetypes

named in H2 is reported descriptively rather than tested against a threshold. A one-at-a-time (tornado) sensitivity analysis shows which parameters drive the results.

Step 7 – Who benefits. The results are mapped onto the distribution of Moroccan households across consumption bands, using data from ONEE, ANRE and the High Commission for Planning (HCP). In 2018, about 80% of ONEE's clients consumed less than 150 kWh per month [23]. This gives the share of households for which a positive NPV is likely under each tariff. Because that breakdown dates from 2018, it is replaced by newer ONEE or HCP data if they are available. The effects on utility revenue and cross-subsidies [19], [20] are discussed in words and not modelled, so that the analysis stays focused on the central question.

Tools and timeline. All calculations are done in Microsoft Excel, with Python (pandas, NumPy) for the Monte Carlo step. The project runs for twelve weeks: tariff, regulatory and billing data (weeks 1–2), PV and demand modelling (3–4), billing engine and validation (5–6), financial model (7–8), Monte Carlo and household mapping (9–10), and writing (11–12).

Ethics. The only household data are electricity bills that households share voluntarily, with written consent. Names, addresses and account numbers are removed before analysis, in line with the Moroccan law on the protection of personal data (Law No. 09-08) [24].

Limitations. (i) Rooftop access: many urban households live in apartment buildings, so the study is limited to houses with private roof terraces. (ii) Calibration sample: at least 20 households is modest, which is why the self-consumption ratio is varied; if fewer can be recruited, the seasonal amplitude becomes a Monte Carlo parameter instead of a calibrated input. (iii) Behavioural response: prosumers may consume more after installation, and a rebound of about 20%, mainly in air-conditioning use at peak hours, has been observed in Austin [25]. (iv) Regulation: low-voltage export remuneration is still unsettled, which is handled through the two export regimes.

Research Topic

Timing is what makes this topic urgent. The implementing decree has been in force only since June 2026, households are making their first investment decisions under it now, and low-voltage export remuneration is still unsettled. It is also new: to the author's knowledge, neither the Moroccan nor the international literature has quantified how whole-bill tariff notches interact with self-consumption, and the counterfactual design isolates that mechanism cleanly. The results would tell households which consumption profiles make solar worthwhile, help ONEE and ANRE anticipate which customers will leave the higher tariff blocks, and show whether capital grants are needed to extend the benefits of Law 82-21 to low-consumption households. The work supports Sustainable Development Goal 7 and Morocco's 52% renewable-capacity target [1].

Quality of Writing

Abbreviations are defined at first use, units are consistent (kWh, kWp, MAD), every figure is sourced, and citations follow IEEE style. The text is written for a reader with basic knowledge of economics and energy.

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